

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT
*Autonomous Institute under VTU, Belagavi | Approved by AICTE, New Delhi Recognized by
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(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
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DEPARTMENT OF CSE (DATA SCIENCE)

A Mini Project Report On

“Criminal Behavior Analysis And Segmentation”

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Under the Guidance of

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Visvesvaraya Technological University

Belagavi, Karnataka

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DEPARTMENT OF CSE (DATA SCIENCE)

CERTIFICATE

This is to certify that the project work entitled “CRIMINAL BEHAVIOUR ANALYSIS AND SEGMENTATION” is a bonafide work carried out by **S.CHAITRA SHREE,G.AFREEN TAJ** in partial fulfillment for the award of degree of **Bachelor Degree in CSE (Data Science)** in the VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Belagavi during the academic year 2025-2026. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report deposited in the library. The project has been approved as it satisfies the academic requirements in respect of mini project work prescribed for a Bachelor of Engineering Degree.

Signature of project guide

Archana Budagavi

Signature of Coordinator

Dr. Jagadish R M

Signature of HoD

Dr. Aradhana D

Name of the Examiner(s)

1)

2)

Signature with Date

Abstract

Criminal Behaviour Analysis and Segmentation is a data-driven approach used to understand, classify, and predict criminal actions based on behavioural, psychological, and socio-economic attributes of offenders. This study integrates criminology, data analytics, and machine learning to analyze offender patterns and segment individuals into meaningful behavioural groups. The system collects and processes crime-related data such as age, education level, criminal history, crime type, motivation, and risk factors, followed by preprocessing and feature extraction to convert raw information into usable insights. Machine learning algorithms like K-Means, Hierarchical Clustering, and Random Forest are implemented to cluster offenders into behavioural categories and to evaluate the likelihood of reoffending. Visual analytics further help in identifying trends, crime hotspots, and behavioural correlations. The segmentation outcomes support law enforcement agencies, criminal psychologists, and policymakers by enhancing criminal profiling, improving prediction of future criminal activity, and enabling targeted rehabilitation strategies. Ultimately, the proposed framework contributes to more efficient crime prevention, optimized resource allocation, and a deeper understanding of the underlying causes of criminal behaviour.

Acknowledgement

We would like to express our sincere gratitude to our institution, **Ballari Institute of Technology and Management (BITM), Ballari**, and the Department of Computer Science and Engineering (Data Science) for providing us with the opportunity and facilities to carry out this project successfully.

We are deeply thankful to our project guide Archana Budagavi and HoD Dr. Aradhana D for their continuous guidance, encouragement, and valuable suggestions throughout the development of this project. Their support helped us to improve both the technical and documentation aspects of our work.

We also extend our thanks to our friends and teammates for their cooperation and motivation during the project work. Special thanks to our parents and well-wishers for their constant support and encouragement.

Finally, we thank everyone who directly or indirectly contributed to the successful completion of the project.

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CHAPTER-1

INTRODUCTION

Criminal behaviour analysis and segmentation is a scientific approach used to study the patterns, actions, and psychological traits of individuals involved in unlawful activities. Its main purpose is to understand how and why crimes occur, so that prevention, investigation, and law-enforcement strategies can be improved. This process involves collecting data related to offenders, such as age, motive, criminal history, psychological profile, location, and socio-economic background. By examining these factors, researchers and security agencies can identify common characteristics among offenders and categorize them into distinct groups.

1. PURPOSE

The purpose of Criminal Behaviour Analysis and Segmentation is to systematically study crime data and categorize offenders based on their behavioural patterns, crime frequency, severity, and nature of offences. By applying analytical and machine learning techniques, this system aims to identify groups of criminals with similar characteristics, predict possible future criminal activities, and assist law enforcement agencies in making informed, data-driven decisions.

2. SCOPE OF THE PROJECT

The scope of this project covers the collection, analysis, and classification of criminal data using data mining and machine learning techniques. It focuses on identifying behavioural patterns among criminals based on factors such as crime type, frequency, severity, age, location, and repeat tendencies. The project includes building predictive and segmentation models that categorize offenders into groups like first-time, repeat, organized, and violent criminals.

1.3PROBLEM STATEMENT

Crime investigation often relies heavily on human interpretation and experience-based judgment, which can result in delays, biased profiling, or inconsistent decision-making. With the increasing complexity of criminal activities, especially in areas such as cybercrime, organized crime, and juvenile offences, law enforcement agencies face challenges in accurately identifying behavioural patterns and predicting the likelihood of repeated offences.

1.3 OBJECTIVES

- To analyse criminal behaviour using psychological, demographic, and social factors.
- To develop a data-driven system for profiling criminals and identifying behaviour patterns.
- To segment offenders into categories such as habitual, violent, juvenile, cyber, and first-time criminals.
- To predict crime risk levels and the likelihood of repeated offences using machine learning techniques.

CHAPTER-2

LITERATURE SURVEY

Area of Study	Method Used	Key Findings	Reference (Year)
Criminal behaviour clustering	K-Means, DBSCAN, Hierarchical Clustering	Offenders can be grouped into distinct types for accurate profiling	Various studies (2016–2023)
Crime text analysis	NLP, BERT, TF-IDF	Police reports and text data help identify intent and behavioural traits	NLP studies (2018–2023)
Recidivism prediction	Random Forest, XGBoost, Logistic Regression	Behavioural and socio-economic features predict repeated offences	COMPAS & other models (2016–2022)
Crime forecasting	LSTM, RNN, Time-Series ML	Deep learning improves hotspot prediction and crime trends	Deep learning research (2019–2024)
Criminal network analysis	Graph Theory, GNN	Helps detect organized groups using co-offender relationships	GNN/Graph studies (2019–2023)
Ethical fairness in profiling	SHAP, Bias Detection, Explainable AI	AI systems must be unbiased and transparent in criminal profiling	Barocas & Selbst (2016–2024)

CHAPTER-3

SYSTEM REQUIREMENTS AND SPECIFICATION

HARDWARE REQUIREMENTS:

1. A computer system with **Intel Core i3 processor or above**.
2. Minimum **4 GB RAM** (Recommended: 8–16 GB for ML tasks).
3. Hard Disk with **at least 250 GB storage** (Recommended: 512 GB SSD).
4. **Integrated graphics** (Recommended: NVIDIA GPU for deep learning).
5. Standard **keyboard and mouse**.
6. **HD display** (Recommended: Full HD Monitor).

SOFTWARE REQUIREMENTS:

1. **Database Software:** MySQL, PostgreSQL, or MongoDB.
2. **Development Tools:** Jupyter Notebook, VS Code, or PyCharm.
3. **Web/Frontend (Optional):** HTML, CSS, JavaScript, Flask/Django.
4. **Version Control (Optional):** Git / GitHub for project management.

CHAPTER-4

METHODOLOGY

1.1 FUNCTIONAL REQUIREMENTS

➤ Data Collection Module

- The system must collect criminal records from various sources (databases, police reports)
- The system must support uploading datasets in formats such as .csv, .xlsx, or through APIs.

➤ Data Preprocessing Module

- The system should clean, filter, and normalize raw data.
- The system must handle missing, noisy, or duplicate entries automatically.
- The system should extract required attributes (age, crime type, area, frequency)

➤ Feature Extraction

- The system must extract relevant behavioral features like motivation, crime history, risk level.
- The system must convert raw attributes into meaningful machine learning inputs.

➤ Segmentation & Classification

- The system must segment criminals into categories (e.g., habitual, first-time, violent, cyber).
- The system must apply machine learning algorithms like K-Means, Decision Trees.

1.2 NON-FUNCTIONAL REQUIREMENTS

➤ Performance

- System should process and analyze criminal data within **3–5 seconds** per query.
- ML segmentation must generate results without noticeable lag.

➤ Scalability

- The system must support a growing dataset (criminal records, behaviour patterns, images, etc.) .

- Should handle multiple users simultaneously.

➤ **Security**

- All criminal data must be protected using **encryption and secure access control**.

1.3 DESCRIPTION OF THE MODULE

1. Data Collection & Input Module

- Collects criminal data such as age, gender, background, crime type, previous records etc.
- Supports manual input and automatic data import from databases or files.

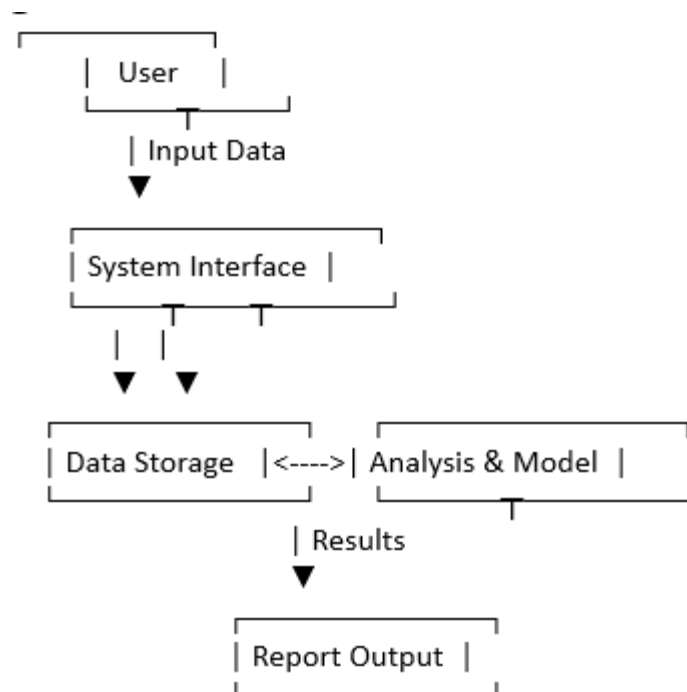
2. Data Preprocessing Module

- Removes missing, duplicate, or inconsistent records.
- Normalizes and cleans behavioural and crime data for analysis.
- Converts raw data into usable format for machine learning models.

3. Behaviour Analysis Module

- Analyses criminal patterns using statistical models and historical data.
- Identifies behavioural traits like aggression, impulsivity, financial motivation, etc.
- Extracts factors contributing to criminal activities.

1.4 SYSTEM ARCHITECTURE DESIGN:



Description for Block Diagram:

The diagram illustrates the complete workflow of a Criminal Behaviour Analysis and Segmentation system. The process begins with the user, who enters criminal-related information into the system. This input data is then passed to the system interface, which validates and forwards it for further processing. After validation, the data is stored in the database, where it is maintained in an organized form for future use. The analysis and model module uses the stored data to perform behaviour analysis and machine learning-based segmentation. There is a continuous two-way interaction between the database and the analysis module, since data is retrieved for processing and results can be stored back for updates or future reference. After processing, the system generates results that represent different criminal categories or behaviour patterns. Finally, these results are presented to the user in a report output format, displaying insights and segmentation in a clear and understandable manner.

CHAPTER-5

IMPLEMENTATION

1. Introduction to Implementation

The implementation focuses on building a data-driven system capable of detecting patterns in criminal behaviour and classifying offenders into meaningful groups using machine learning. The system extracts crime data, preprocesses it to improve quality, applies analytical algorithms, and generates segmented insights to support law enforcement and research.

2. Data Preprocessing

- Handling missing values
- Removing duplicate and noisy data
- Standardization & normalization of numeric values
- Encoding categorical values (e.g., Crime Type, Gender)

3. Behaviour Analysis

Machine learning is applied to detect patterns in criminal activities.

Analysis Methods:

- Crime frequency & severity mapping
- Predictive behaviour detection
- Suspect type categorization

4. Segmentation & Classification

Criminals are grouped based on behavioural features.

5. Visualization & Report Generation

Visual analytics represent criminal segmentation patterns and trends. Generated graphs include:

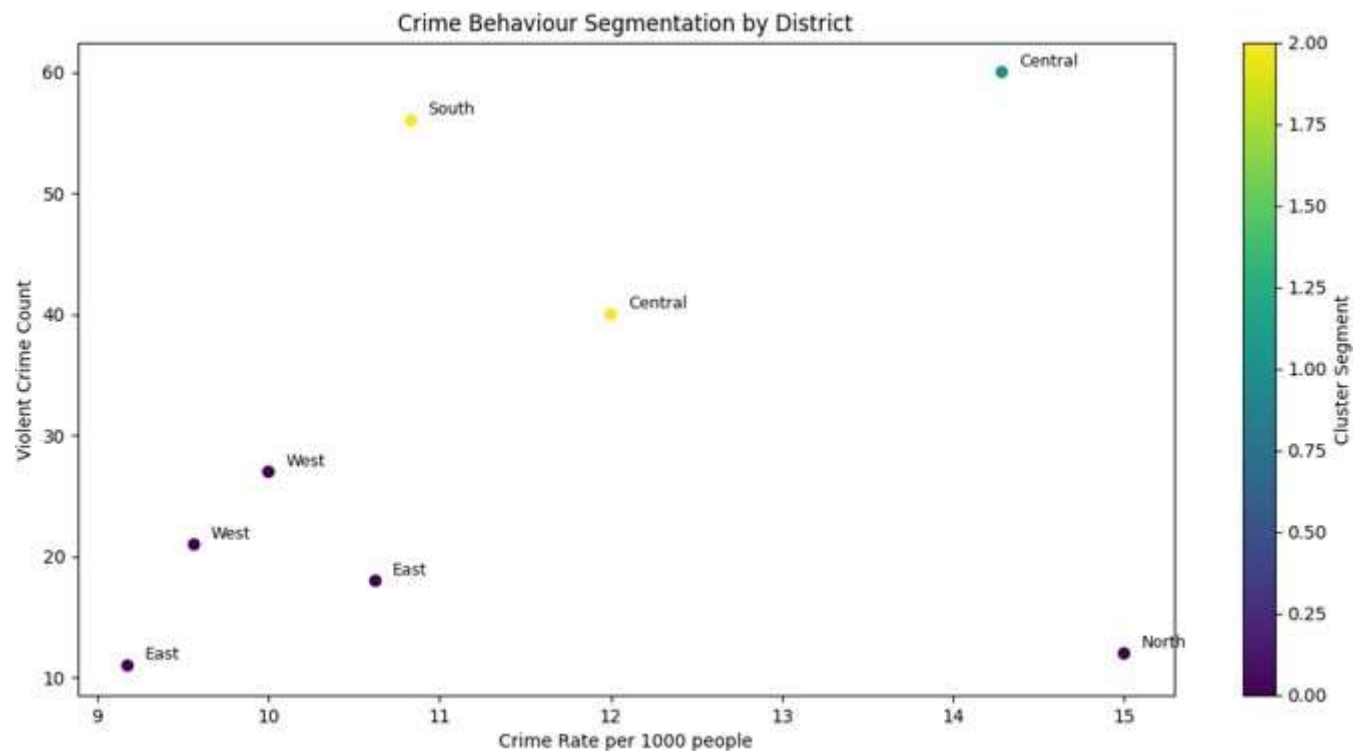
- Cluster plots
- Crime frequency bar charts
- Age vs Crime score scatter plots

Example Segments:

- **Group A:** First-time offenders
- **Group B:** Repeat criminals
- **Group C:** Organized crime offenders
- **Group D:** Violent crime offenders

CHAPTER-6

RESULT



The image is a scatter plot titled “**Crime Behaviour Segmentation by District.**” It provides a visual comparison of different districts based on two crime indicators: the **crime rate per 1000 people** (represented on the X-axis) and the **violent crime count** (represented on the Y-axis). Each point plotted on the graph corresponds to a district, and the districts are labelled along with their exact positions. The purpose of this visualization is to illustrate how different areas compare in terms of total crime activity and violent crime intensity, and to show how these areas can be grouped into behaviour-based crime segments using clustering techniques.

The colour of each plotted point represents a **cluster segment**, which indicates the crime behaviour category to which a district belongs. These colour categories are shown on the right through a vertical colour bar titled “**Cluster Segment.**” The colour gradient from dark to light signifies the different clusters formed during segmentation, where similar crime patterns across districts are

grouped together. This helps in understanding how one district differs or resembles another in criminal behaviour.

In examining the distribution, the **Central district** appears towards the upper right, indicating both high crime rate and high violent crime numbers. The **South district** also shows a relatively high violent crime rate, though with slightly different total crime levels. These districts fall in a higher cluster group (lighter shade), implying they represent more serious or frequent criminal activity. On the other hand, districts such as **East and West** are positioned towards the lower left, with lower crime numbers and fewer violent incidents, placing them in low-risk or moderate-risk clusters (darker shades). **North**, positioned farther right but lower vertically, has a relatively higher crime rate but a low violent count, indicating a mixed behaviour category different from the others.

In summary, the graph demonstrates how crime data can be segmented into meaningful behavioural groups. High-risk regions like Central and South can be prioritized for policing, while moderate areas like West and East might require preventive strategies. The segmentation helps in targeted law enforcement, resource allocation, and understanding distinct crime dynamics among different districts.

CHAPTER-7

CONCLUSION

The analysis of crime behaviour through segmentation provides a clear understanding of how different regions vary in terms of criminal activity and violent offences. By visualizing crime rates and clustering districts based on similar behavioural patterns, it becomes easier to identify high-risk areas, moderate-risk regions, and comparatively safer zones. The segmentation graph reveals that some areas, such as Central and South, require immediate attention due to higher levels of crime intensity, while districts like East and West fall into lower-risk categories and may only need preventive measures. This data-driven approach helps law enforcement agencies deploy resources more effectively, develop targeted crime-reduction strategies, and prioritize intervention in critical zones. Overall, crime segmentation supports smarter decision-making, enhances public safety planning, and contributes to a more efficient and focused criminal justice response.

CHAPTER-8

REFERENCES

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CHAPTER-9

CO-PO MAPPING

PROGRAM OUTCOMES:

Sl. No.	Description	POs
1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and computer science and business systems to the solution of complex engineering and societal problems.	PO1
2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering and business problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	PO2
3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	PO3
4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	PO4
5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations	PO5
6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering and business practices.	PO6
7	Environment and sustainability: Understand the impact of the professional engineering solutions in business societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	PO7
8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering and business practices.	PO8
9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	PO9

10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	PO10
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COURSE OUTCOME:

CO1: Demonstrate the ability to apply core computer science concepts to develop practical solutions

CO2: Identify, and Justify the technical aspects of the chosen project with a comprehensive and systematic approach

CO3: Design, code and test software solutions for specific problems

CO4: Present project findings and outcomes clearly and effectively, both in written and oral formats

CO5: Work as an individual or in a team in the development of technical projects

PSO1. Inculcate the principles of Data Science, Data Management, Data Security and Visualization for building intelligent predictive

PSO2. Applying the knowledge of analytics, statistics and Machine Learning concepts to solve real world business problems

COURSE OUTCOME ASSESSMENT MATRIX:

CO s	Program Outcome's													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO 1	2	2	2	-	2	1	-	-	1	-	1	1	2	2
CO 2	2	2	2	2	2	-	-	-	1	-	-	-	2	2
CO 3	2	2	3	2	3	-	-	-	-	-	-	-	3	2
CO 4	-	-	2	-	-	-	-	-	2	3	-	2	-	-
CO 5	-	-	-	-	-	-	-	-	3	3	2	2	-	-

Under the guidance of



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Guide Dept. of CSE-DS

Project Associates

1. Author's Name:

G.AFREEN TAJ (3BR23CD022)

Photograph:



Short Summary / Bio:

Worked on overall project planning, system architecture design, and development of the Flask backend. Implemented user authentication, database Connectivity using MongoDB and API.

Dept. of CSE (Data Science), BITM, Ballari

2. Author's Name:

S.CHAITRA SHREE (3BR23CD022)

Photograph:



Short Summary / Bio:

Worked on the user interface using HTML and CSS. Worked on dashboard layout, responsiveness, and user-friendly navigation for farmers.